ES-DETR: Edge-Guided State-Space DETR for Foggy Remote Sensing Object Detection

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***Abstract*—Object detection in optical remote sensing imagery is severely affected by adverse weather conditions, such as fog and haze, which degrade image quality and obscure structural details. Although recent Transformer-based detectors have achieved promising performance, they suffer from quadratic computational complexity for high-resolution inputs and tend to produce imprecise object boundaries in degraded scenes. To address these issues, we propose an Edge-Guided State-Space DETR (ES-DETR), an end-to-end detection framework that integrates linear-complexity state-space modeling with structural priors. Specifically, a Laplacian Edge-Aware Module (LEM) is designed to extract high-frequency boundary information from foggy images. Moreover, a Structural-Prior-Driven Mamba Fusion Module (SMF) is introduced to incorporate edge-derived structural priors into the Mamba architecture for long-range dependency modeling and feature fusion. This design effectively restores degraded semantic representations. Extensive experiments on foggy remote sensing benchmarks demonstrate that the proposed ES-DETR outperforms state-of-the-art detectors while maintaining a favorable accuracy–efficiency trade-off.**

***Index Terms*—Object detection, remote sensing image, state-space model, degradation-aware learning, structural prior.**

I. Introduction

Optical remote sensing imagery plays a crucial role in various Earth observation applications [1], [2], [3], [4]. However, real-world images are often degraded by adverse weather conditions, such as fog and haze [5]. These atmospheric phenomena cause severe contrast reduction and color shifts, obscuring fine-grained structural details and object boundaries. Consequently, accurate object detection in foggy remote sensing scenarios remains a formidable challenge.

Existing deep learning-based detectors struggle to balance performance and efficiency in such degraded scenes. While Convolutional Neural Networks (CNNs) [6], [7], [8], [9], [10], [11] are efficient, their localized receptive fields limit their ability to capture the global context necessary for recovering heavily obscured targets. Recently, Transformer-based architectures [12], [13], [14], [15], [16] particularly DETR variants [17], [18], [19] have achieved remarkable success by leveraging global self-attention. Unfortunately, the quadratic computational complexity of Transformers makes them computationally prohibitive for high-resolution remote sensing images. It is worth noting that State-Space Models (SSMs) [20], [21], [22], [23] achieve global receptive fields with linear complexity. However, directly applying standard SSMs to foggy images is suboptimal. Atmospheric scattering predominantly attenuates high-frequency components while preserving low-frequency appearance. Therefore, explicit structural priors are required to recover these vulnerable degraded boundaries, rather than relying on general-purpose enhancements, to ensure accurate localization for remote sensing targets.

To address these limitations, we propose an end-to-end Edge-Guided State-Space DETR framework for foggy remote sensing object detection. Unlike conventional edge enhancement, our motivation is to extract degradation-resistant high-frequency responses as structural priors for linear-complexity sequence modeling. Specifically, the proposed Laplacian Edge-Aware Module (LEM) does not output generic edge maps, but constructs explicit structural priors to characterize target geometries robustly under atmospheric scattering. Instead of standard feature fusion, we propose a Structural-Prior-Driven Mamba Fusion Module (SMF). It embeds these priors into selective state-space evolution, enabling boundary-aware dynamic representation learning rather than the generic sequence modeling of existing Mamba detectors.

The main contributions are summarized as follows.

- We propose edge-guided state-space DETR, which introduces state-space models into foggy remote sensing object detection and effectively alleviates the computation–performance trade-off.
- We design a Laplacian edge-aware module, which explicitly extracts high-frequency boundary cues from foggy images, thereby compensating for structural degradation caused by adverse weather conditions.

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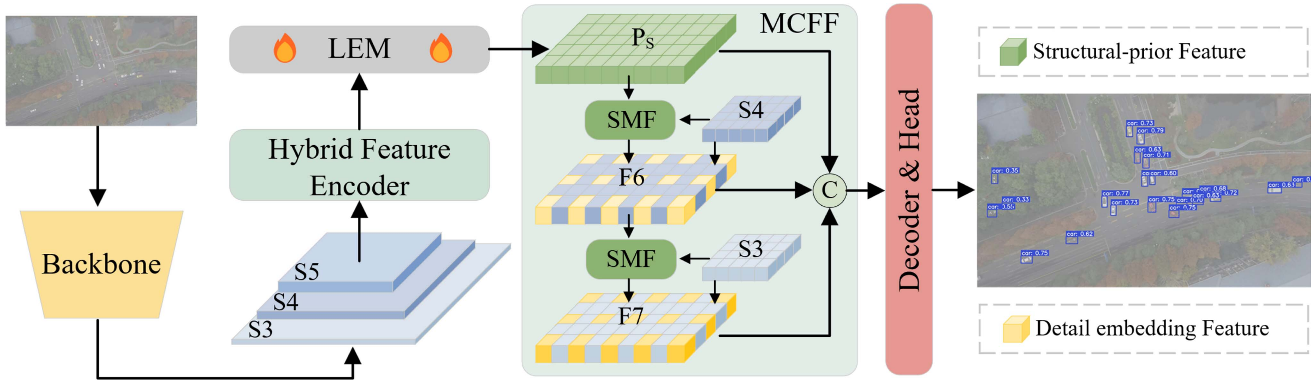


Fig. 1. Overview of the proposed network. We first feed the features from the last three stages of the backbone (S3, S4, S5) into the Hybrid Feature Encoder. Subsequently, LEM processes these representations to generate the top-level structural-prior feature ($\mathcal{P}_s$). Following this, the Mamba-driven Cross-scale Feature Fusion (MCFF) module efficiently integrates the multi-scale features. Specifically, it utilizes SMF blocks to fuse the high-level structural-prior feature with shallower detail embedding features (S4, S3) from the backbone. Finally, the concatenated hierarchical features (F6, F7) are fed into the Decoder Head to iteratively optimize object queries and generate final categories and bounding boxes.

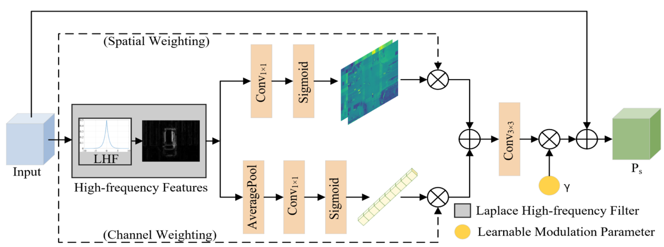


Fig. 2. Illustration of the LEM. It captures boundary-aware structural priors via a Laplacian high-frequency filter. These features are adaptively recalibrated through spatial and channel weighting branches, then modulated by a learnable parameter γ to enhance object details under adverse weather.

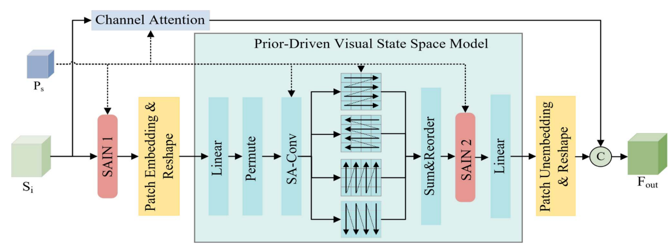


Fig. 3. Illustration of the SMF module. A prior-driven state-space model fuses input features ($\mathbf{S}_i$) and structural priors ($\mathcal{P}_s$) via hidden-state modulation. Residual integration with channel attention and Structural-Aware Instance Normalization (SAIN) layers yields the output $\mathbf{F}_{\text{out}}$.

- We introduce a structural-prior-driven Mamba fusion module, which leverages the extracted edge information as structural priors to guide the Mamba architecture, enabling deep and boundary-aware feature fusion.

II. Methodology

A. Network Overview

The proposed edge-guided state-space DETR (Fig. 1) mitigates structural degradation under foggy conditions. Given a foggy remote sensing image $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$, HGNetv2 is adopted as the backbone to extract multi-scale features $\mathbf{S}_{i\in\{3,4,5\}}$ from its last three stages. The LEM (Fig. 2) then derives structural priors $\mathcal{P}_s$ from deep features. The structural-prior-driven Mamba fusion module (Fig. 3) jointly models $\mathbf{S}_i$ and $\mathcal{P}_s$, using the prior to guide linear-complexity state-space modeling. The fused features are then fed into a standard DETR decoder. All components, including the LEM and SMF, are jointly optimized end-to-end.

B. Laplacian Edge-Aware Module

Atmospheric scattering predominantly acts as an anisotropic low-pass filter, severely attenuating the high-frequency topological invariants of remote sensing targets. To rigorously extract boundary-sensitive cues, we formulate the Laplacian edge-aware module as a topological differential operator. To construct an adaptive structural prior, we first compute the continuous high-frequency response manifold via the trace of the Hessian matrix. We then subject it to a dual-space orthogonal decomposition, synthesizing the refined topological invariant $\tilde{\mathcal{M}}_{hf}$. This process can be formalized as

$$\begin{cases} \tilde{\mathcal{M}}_{hf} &= (\mathcal{M}_{hf} \otimes \mu_s) \oplus (\mathcal{M}_{hf} \otimes \mu_c) \\ \mathcal{M}_{hf} &= \operatorname{tr}(\mathcal{H}(\mathbf{I})) \cong \mathbf{K}_{lap} \circledast \mathbf{I} \end{cases}, \tag{1}$$

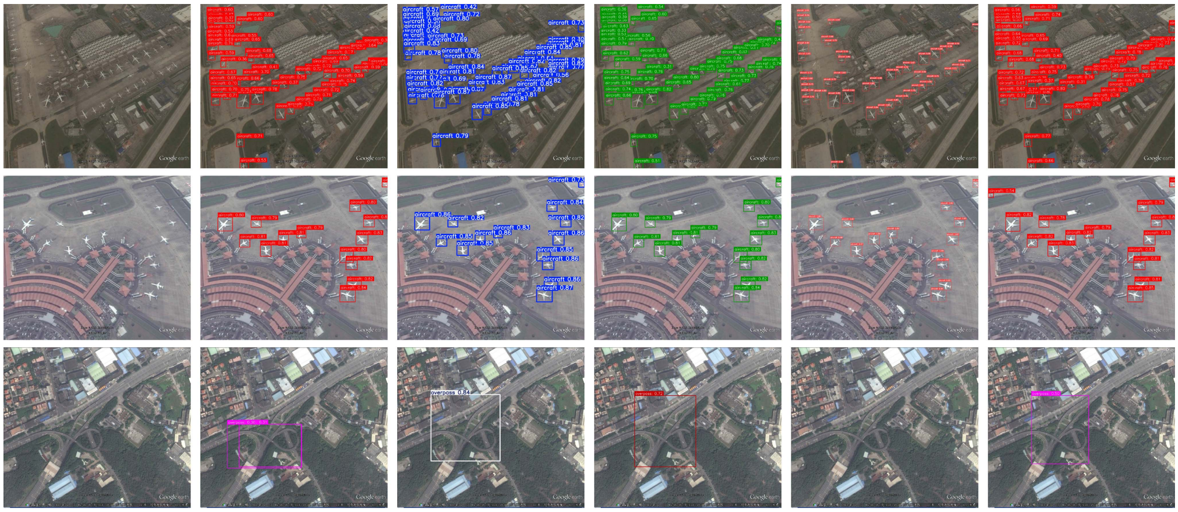
where $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$ is the degraded image, μ_s and μ_c represent the spatial and channel-wise Lebesgue measures respectively, and $\circledast$ denotes spatial convolution. Finally, as shown in Fig. 2, the explicit structural prior $\mathcal{P}_s$ is constructed by injecting this modulated invariant back into the spatial domain, which can be represented as

$$\mathcal{P}_s = \mathbf{I} + \gamma \cdot \Psi_{agg}\left(\tilde{\mathcal{M}}_{hf}\right). \tag{2}$$

where $\otimes$ and $\oplus$ denote element-wise multiplication and addition, γ is a learnable spectrum scaling parameter, and Ψ_{agg} acts as a dense neighborhood aggregation functional. This formulation effectively recalibrates the differential energy of object contours.

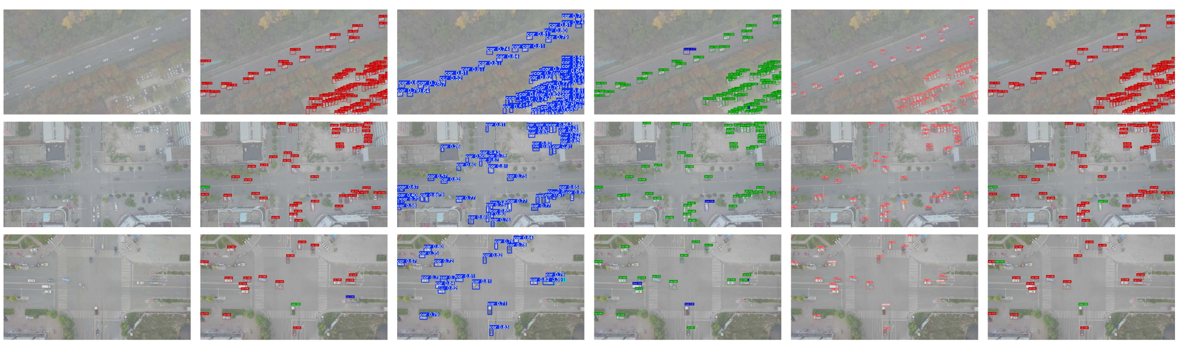
C. Prior-Driven Selective State-Space Fusion

While standard discrete State-Space Models (SSMs) provide linear complexity, they inherently constitute a time-invariant dynamical system lacking boundary awareness. To bridge this



(a) Input (b) DEIM [16] (c) YOLO26 [24] (d) D-FINE [14] (e) FFCANet [25] (f) Ours

Fig. 4. Qualitative comparison with representative methods on the RSOD [25] dataset.



(a) Input (b) DEIM [16] (c) YOLO26 [24] (d) D-FINE [14] (e) FFCANet [25] (f) Ours

Fig. 5. Qualitative comparison with representative methods on the HazyDet [6] dataset.

gap, the proposed structural-prior-driven Mamba fusion intrinsically elevates the sequence modeling into a selective, prior-conditioned control system.

As illustrated in Fig. 3, let the input multi-scale feature sequence be $\mathbf{S}_i(t)$ and the corresponding structural prior be $\mathcal{P}_s(t)$. The continuous-time state evolution is governed by a prior-conditioned Ordinary Differential Equation (ODE), which can be formulated as

$$\frac{d\mathbf{h}(t)}{dt} = \mathbf{A}\mathbf{h}(t) + \mathbf{B}\left(\mathcal{P}_s(t)\right)\cdot \mathcal{F}_{\text{SAIN}}\left(\mathbf{S}_i(t) \mid \mathcal{P}_s(t)\right), \quad (3)$$

where $\mathcal{F}_{\text{SAIN}}$ represents the affine transformation parameterized by the structural prior via Structural-Aware Instance Normalization. Through the Zero-Order Hold (ZOH) discretization, the local step size Δ_k and projection matrices are transformed into data-dependent selective filters, which can be expressed as

$$\bar{\mathbf{A}}_k = \exp(\Delta_k \mathbf{A}), \quad \bar{\mathbf{B}}_k = \int_0^{\Delta_k} \exp(\tau\mathbf{A})\mathbf{B}(\mathcal{P}_{s,k})\, d\tau, \quad (4)$$

Finally, the discrete-time selective transition and semantic integration are synthesized into a unified fusion space. By employing a global attention functional $\mathcal{G}_{\text{attn}}$ on the residual connection, the output feature is recursively formulated as

$$\mathbf{F}_{\text{out}} = \left(\mathbf{S}_{i,k} \odot \mathcal{G}_{\text{attn}}(\mathbf{S}_{i,k})\right) \,\|\, \mathbf{C}_k\left(\bar{\mathbf{A}}_k\mathbf{h}_{k-1} + \bar{\mathbf{B}}_k\hat{\mathbf{S}}_{i,k}\right). \quad (5)$$

where $\hat{\mathbf{S}}_{i,k}$ is the discrete modulated input and $\|$ denotes tensor concatenation. Unlike simple concatenation, this modulation strategy mathematically forces the dynamic system to allocate higher spectral energy along the degraded boundaries extracted by $\mathcal{P}_s$, achieving robust prior-guided fusion.

III. Experiment

A. Dataset and Implementation Details

Datasets: The proposed method and all competing methods are evaluated on the HazyDet [6] and RSOD [25] datasets under the same settings. The HazyDet [6] dataset is a UAV-based benchmark for object detection under haze conditions, including both synthetic and real haze images with three vehicle categories. The synthetic subset contains approximately 11,000 images and 365,000 instances, with a train/val/test split of 8:1:2, while the real subset comprises 600 images for evaluating robustness under degraded visibility. The RSOD [25] dataset consists of 976 high-resolution remote sensing images with four object categories. It is characterized by small and densely distributed targets, posing significant challenges for object detection and feature representation.

Implementation Details: All experiments are conducted on a machine equipped with an NVIDIA RTX 4060 Ti GPU (16 GB), an Intel Core i5-13600KF CPU, and 32 GB RAM. All models are

TABLE I
QUANTITATIVE COMPARISON WITH REPRESENTATIVE METHODS ON HAZYDET [6] AND RSOD [25]. BEST RESULTS IN RED, SECOND-BEST IN BLUE.

Methods	Pub'Year	HazyDet [6]						RSOD [25]						Params(M)↓	GFLOPs↓	FPS↑
		AP↑	AP_{50}↑	AP_{75}↑	AP_S↑	AP_M↑	AP_L↑	AP↑	AP_{50}↑	AP_{75}↑	AP_S↑	AP_M↑	AP_L↑			
DETR [12]	ICCV&21	46.8	66.3	54.0	13.0	47.0	67.9	63.5	91.0	70.5	43.0	68.5	66.0	41.59	86.21	45
DINO [15]	ICLR&23	47.9	67.8	54.8	13.5	48.2	69.5	66.5	93.5	75.0	44.5	71.0	70.5	47.29	128.64	35
YOLOv7 [26]	CVPR&23	46.7	66.0	53.1	12.5	46.5	67.5	63.7	90.2	71.1	43.4	68.9	66.1	36.95	104.73	85
YOLOv9s [27]	ECCV&24	47.5	67.5	54.5	12.8	48.0	69.3	64.0	91.5	71.0	44.0	69.5	67.5	7.21	26.76	155
YOLOv10s [28]	NIPS&24	47.8	68.2	54.9	13.0	48.5	69.8	64.3	92.0	71.5	44.5	70.0	68.0	7.20	21.67	158
YOLO11s [29]	2024	48.3	68.9	55.3	12.9	49.5	70.9	64.7	92.4	72.3	45.2	70.7	68.6	9.43	21.58	161
RT-DETR [13]	CVPR&24	48.1	68.5	55.0	13.2	48.8	70.5	66.8	93.8	76.5	44.8	71.5	71.0	42.00	152.63	55
FFCANet [25]	TGRS&24	47.2	67.1	54.1	13.3	47.5	68.5	65.5	92.5	73.0	45.0	71.0	69.5	7.15	51.27	110
DHCNet [6]	TGRS&25	47.6	67.8	54.6	13.5	48.0	69.2	66.0	93.2	74.0	45.2	71.8	70.0	18.45	55.30	95
MASFNet [30]	TGRS&25	47.0	66.8	54.2	13.1	47.3	68.2	65.2	92.2	72.5	44.6	70.5	68.8	6.02	10.71	170
D-FINE [14]	ICLR&25	48.5	69.0	55.5	13.8	49.2	71.2	67.5	94.0	78.5	45.1	72.8	72.0	10.41	25.07	145
DEIM [16]	CVPR&25	47.3	67.0	54.5	12.8	47.6	68.8	68.2	94.5	79.7	44.1	73.4	72.6	20.36	60.25	102
CID [31]	KBS&25	46.0	65.5	53.0	12.0	46.0	66.5	64.5	91.8	71.8	43.5	69.0	67.8	82.58	324.48	25
MFPD [32]	SPL&26	47.4	67.2	54.7	13.2	47.8	69.0	66.2	93.3	74.5	45.0	72.0	70.2	33.23	50.61	65
YOLO26s [24]	2026	48.6	69.1	55.6	13.6	49.4	71.0	65.9	93.0	73.5	46.0	71.2	69.8	9.62	20.83	150
DFFR [18]	TII&26	47.7	67.5	55.8	13.4	48.1	69.4	66.5	93.6	75.2	45.1	72.3	70.8	31.82	100.17	62
Ours	-	**48.9**	**69.3**	**55.6**	**14.1**	**49.7**	**71.5**	**69.4**	**96.7**	**82.4**	**45.6**	**73.4**	**73.9**	**6.52**	**7.19**	**185**

trained from scratch for 300 epochs. AdamW is used with a base learning rate of 8×10^{-4}, while the backbone learning rate is set to 4×10^{-4}. The training batch size is 16. Efficiency metrics (GFLOPs and FPS) are evaluated at 640×640 resolution with a batch size of 1.

B. Evaluation Metrics and Comparison Methods

Evaluation Metrics: Detection performance is evaluated using Average Precision (AP) metrics, including AP, AP_{50}, AP_{75}, and scale-specific metrics (AP_S, AP_M, AP_L).

Comparison Methods: We compare the proposed method with a series of representative object detectors, including Transformer-based methods such as DETR [12], DINO [15], and DEIM [16], as well as CNN-based detectors from the YOLO family, including YOLOv7 [26], YOLOv9 [27], YOLOv10 [28], YOLOv11 [29], and YOLO26s [24]. In addition, several recent advanced detectors are included for comprehensive comparison, such as RT-DETR [13], FFCANet [25], DHCNet [6], MASFNet [30], D-FINE [14], CID [31], MFPD [32], and DFFR [18].

C. Qualitative Comparison

Figs. 4 and 5 show qualitative comparisons on the RSOD [25] and HazyDet [6] datasets. Under hazy conditions, the proposed method yields more complete and accurate detections for both dense small objects and sparse large objects. In complex scenes with densely distributed small objects, existing methods (e.g., DEIM [16] and YOLO26 [24]) often suffer from missed detections or imprecise localization. By explicitly modeling edge information, the proposed method better preserves object boundaries, leading to improved localization and fewer false positives. Under hazy conditions, the proposed method leverages structural priors to guide feature aggregation, enabling more reliable detection of weak targets and improved robustness to degradation.

D. Quantitative Comparison

Table I reports the quantitative results on the HazyDet [6] and RSOD [25] datasets. The proposed method achieves the best overall performance across most evaluation metrics. On the HazyDet [6] dataset, ES-DETR attains the highest AP, AP_{50}, AP_S, AP_M, and AP_L, demonstrating its effectiveness in degraded scenes. It also achieves superior performance on the RSOD [25] dataset in terms of AP and AP_{50}, with consistent improvements across object scales. Furthermore, the proposed method achieves a favorable balance between accuracy and efficiency, with only 6.52 M parameters and 7.19 GFLOPs while maintaining the highest inference speed.

TABLE II
ABLATION RESULTS OF DIFFERENT COMPONENTS ON THE RSOD DATASET [25]. BEST RESULTS ARE IN BOLD.

Case	LEM	SMF	AP↑	AP_{50} ↑	AP_{75} ↑	Params↓	GFLOPs↓	FPS↑
0			68.0	94.1	79.5	**3.72M**	7.12G	**215**
1	✓		68.6	95.3	80.9	3.89M	7.24G	198
2		✓	68.8	95.6	81.2	6.28M	**7.07G**	203
3	✓	✓	**69.4**	**96.7**	**82.4**	6.52M	7.19G	185

E. Ablation Study

Ablation experiments are conducted on the RSOD [25] dataset. As shown in Table II, introducing LEM improves AP from 68.0 to 68.6, with consistent gains in AP_{50} and AP_{75}. Using only the SMF module achieves 68.8 AP with lower computational cost (7.07 GFLOPs), indicating improved efficiency. Combining LEM and SMF yields the best performance, achieving 69.4 AP, 96.7 AP_{50}, and 82.4 AP_{75}, confirming their complementarity. Despite the increased complexity, the model maintains high efficiency at 185 FPS.

IV. CONCLUSION

In this work, we propose an edge-guided state-space DETR for object detection in degraded remote sensing scenes. The key idea is to incorporate explicit structural priors into efficient state-space modeling to enhance feature representation under adverse conditions. Specifically, a Laplacian edge-aware module is designed to extract high-frequency boundary information, while a structural-prior-driven Mamba fusion module is introduced to guide long-range dependency modeling with edge-aware cues. These designs jointly improve both localization accuracy and robustness in challenging scenarios. Extensive experiments on the HazyDet and RSOD datasets demonstrate that the proposed method consistently outperforms representative detectors in terms of accuracy, while maintaining competitive computational efficiency. In future work, we will explore more lightweight architectures and broader remote sensing applications.

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